Alchip

3661.TW | TWSE | 2026-10-07

Q4 2027 model ceiling: TWD 6,067.92

Analytical Base 44/60 = 73 | Watch

20x forward-four-quarter modeled IFRS operating profit, plus only excess treasury after collateral/operating reserves, less lease and minority claims, divided by matched forward diluted shares

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

Snapshot: TWD 4,180.00 on 2026-10-06 close. Model diluted shares at Q3 2026: 87.766m. Indicative diluted capitalization: TWD 366,861.9m. Quote and share dates/bases may differ; legal shares are reconciled in the analysis.

All operating values are in TWD m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	25,442.8	20x	88.429	6,067.92

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

Company operating profit and matching common-share claims. 20x forward-four-quarter modeled IFRS operating profit, plus only excess treasury after collateral/operating reserves, less lease and minority claims, divided by matched forward diluted shares

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use TWD m unless a different unit is stated.

Thesis

Alchip earns its place in the AI supply chain by turning a customer's chip architecture into working, manufacturable silicon. Its opportunity is unusually large because customers want customized compute while advanced-node implementation, package routing, testing and production control become harder. The investment case nevertheless depends on profit per completed program and cash conversion. A much larger chip bill passing through the accounts does not automatically create a similarly larger intellectual-property business.

The operating model uses a 20x forward-four-quarter IFRS operating-profit ceiling, with only genuinely excess treasury added after collateral and operating reserves. The Q4 2027 ceiling is approximately NT$6,068 per share. The valuation window is Q4 2027 through Q3 2028, and the matching modeled diluted denominator is 88.43 million shares. The analytical Base is 73, a Watch classification: strong engineering relevance and visible shipment acceleration are offset by customer concentration, manufacturing dependence and unresolved program economics. These are model outputs, not company guidance or a bank target.

What the customer buys and what Alchip books

The issuer's engagement map offers several entry points. A customer can hand off a specification, RTL or netlist, completed physical design, packaged chip, or an established production program. Alchip's work can therefore span architecture support, physical implementation, design for test, mask preparation, packaging, qualification, yield improvement and production logistics. The customer may retain its architecture and software while outsourcing a particularly demanding part of physical realization. [business_models; FY2025_annual, pp.98-111]

This flexibility also makes revenue comparisons dangerous. The 2025 annual report says its broad ASIC and chip-production category includes non-recurring engineering activity as well as mass-production management. The separately labeled NRE or commissioned design-service category is much narrower. In H1 2026, the statutory statement shows USD366.596 million of ASIC/wafer sales and USD7.498 million of standalone design-service revenue. It would be incorrect to conclude that all engineering economics were only USD7.5 million, or to label a much larger assumed NRE pool as a reported fact. [Q2_2026_financials, pp.41-42; FY2025_annual, p.98]

The model consequently uses three expressly assumed economic allocations: engineering-rich design/prototype/program work, leading-edge North American production, and other production. These buckets are not audited segments. They let the analysis distinguish high-value work from manufacturing pass-through without pretending that the company has published product-level margins. Gross-versus-net recognition must follow control, responsibility and contractual economics; merely seeing the word consignment in a technical description does not establish that every major accelerator order is booked net.

An illustrative contract transition explains the sensitivity. If a USD100 manufacturing bill produces USD10 of gross profit, a change to net fee presentation could reduce reported revenue toward USD10 while preserving much of that gross profit. Applying a sales multiple before and after that change would invent an economic collapse. This operating-profit model instead watches the retained fee, direct costs and working capital. Customer-owned wafers

or memory are not inserted into Alchip's revenue without evidence that Alchip is the principal.

IV. Overview of Business Operations

4.1 Business Activities

4.1.1 Business scope

A. Main business operations

Alchip is a leading ASIC (Application Specific Integrated Circuit) service company specializing in High-Performance Computing (HPC) and AI (Artificial Intelligence) infrastructure applications.

B. Revenue by service category

Unit: NT$ thousand

Category	2024		2025	
	Amount	%	Amount	%
ASIC and Chip production	51,586,295	99.27	30,446,828	98.45
NRE	365,751	0.70	446,352	1.44
Others	16,524	0.03	32,912	0.11
Total	51,968,570	100.00	30,926,092	100.00

C. Main products and services

(1) ASIC and Chip production: Provides customers with Non-Recurring Engineering (NRE) of Application Specific Integrated Circuits (ASICs) and System on Chips (SoCs); and mass production management of wafer manufacturing, packaging, and testing.

(2) Non-Recurring Engineering (NRE): Provides the circuit design component database and Silicon Intellectual Property (SIP) required by product design to produce circuit diagrams for the mask-making process, manufacture masks, wafer, cutting, and packaging on a consign basis; then delivering trial production samples after product testing.

(3) Others: Provides customers with one-off back-end wafer fabrication, packaging, and testing.

D. New products (services) development

The Company is committed to the research, development, and manufacturing of leading-edge integrated circuits, including custom design utilities and design methodologies that ensure quality services. It also provides a chiplet technology platform; an HPC IP portfolio (including HBM, UCIe, PCIe and SerDes); and the latest 3.5D heterogeneous packaging capabilities to meet AI/HPC market demands.

4.1.2 Industry overview

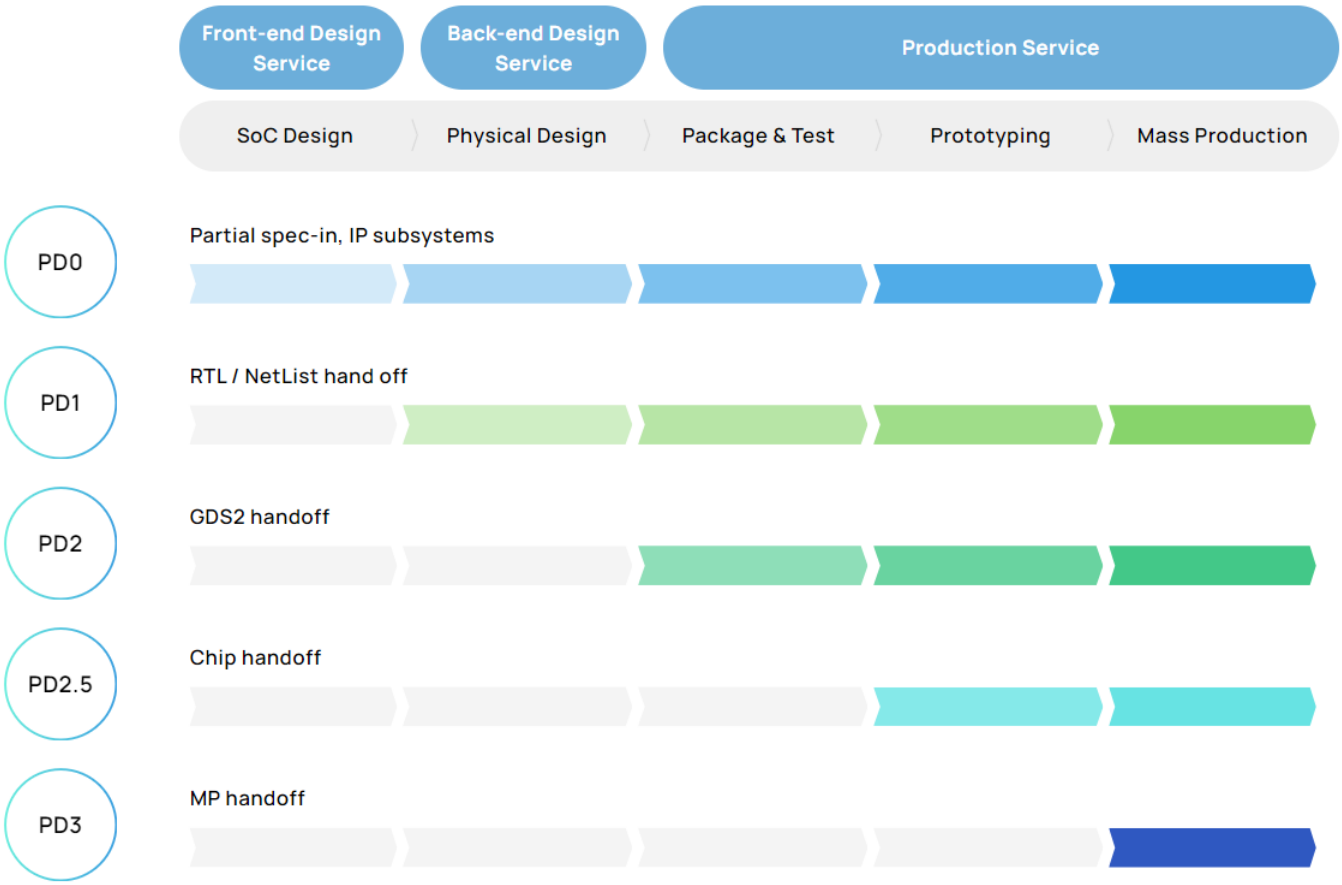
A. Current status and industry outlook

As Moore's Law approaches its physical limits, the continued performance scaling of general-purpose processors faces increasing challenges. By contrast, Application-Specific Integrated Circuits (ASICs) are gaining prominence due to their highly optimized architectures for specific workloads and algorithms. By offering enhanced computational throughput and significantly improved power efficiency, ASICs are a critical enabler of innovation and performance breakthroughs across the semiconductor industry. Alchip provides design and manufacturing services for highly complex ASICs and SoCs. The industry's current status and outlook are as follows:

(1) Rise of fabless ASIC

In the past, a system company could choose to develop its own ASICs and/or SoCs to create differentiation and stay competitive; or to entrust the turnkey production process, including design,

Original issuer exhibit | The statutory product category includes NRE-related activity; the small separately reported design-service line is not the complete economic engineering wallet.



Original issuer exhibit | Issuer engagement map: the customer can hand off at specification, RTL, GDS, packaged-chip or mass-production stages. Economic capture varies with the hand-off.

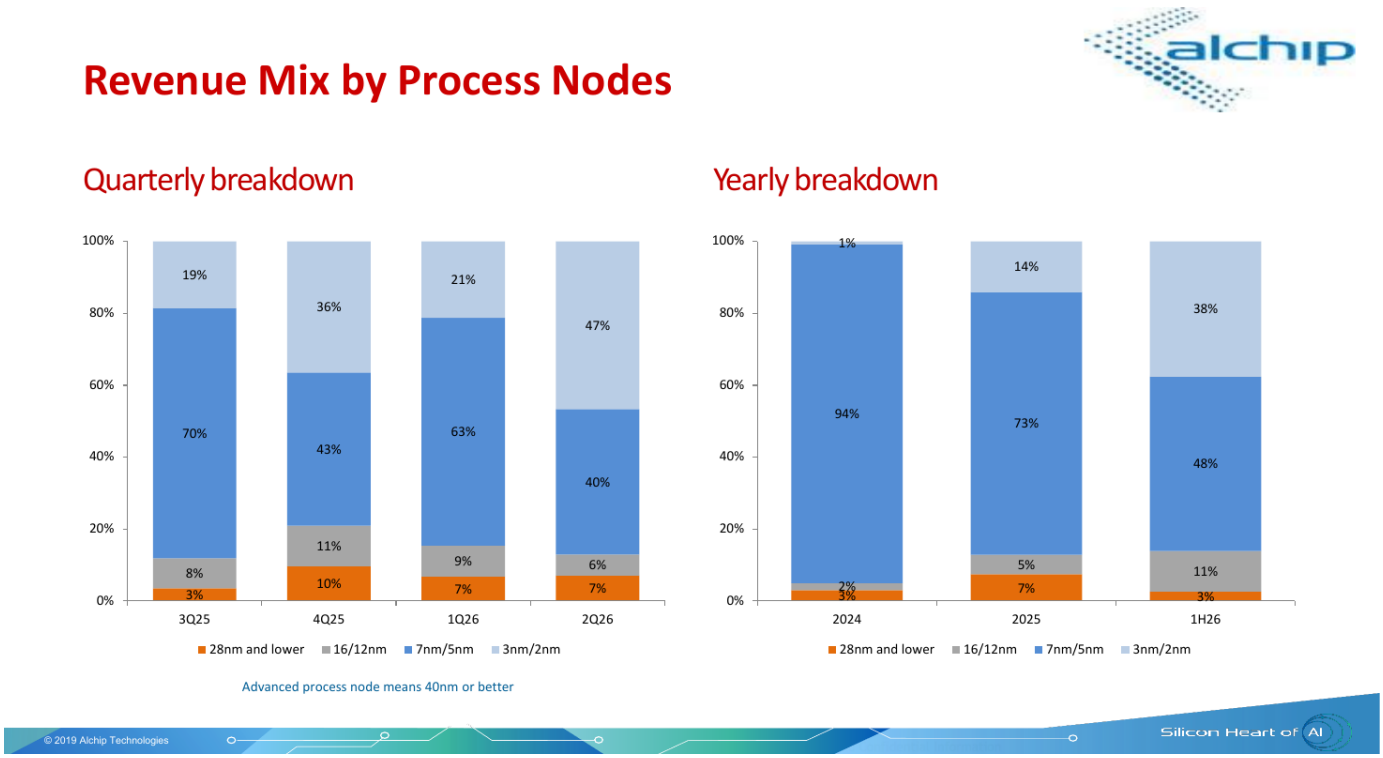
The engineering bottleneck and its limits

At 3nm and 2nm, physical design requires more than shrinking a previous layout. Routing, timing closure, power delivery, thermal density, test coverage and package escape routing interact. Chiplets introduce another set of interfaces, synchronization and assembly dependencies. Alchip's March announcement describes a 2nm platform, 3nm/5nm I/O-chiplet support, multiple CoWoS approaches and SoIC-related work. A successful test chip validates part of the engineering process; it is not proof of a customer's full-reticle production yield or an allocated shipment volume. [2nm]

The September Synopsys collaboration is useful technical evidence. It combines Synopsys 224G PHY IP with Alchip's signal/power-integrity and packaging work, addressing crowded die edges and package breakout. It supports the proposition that physical implementation is valuable as bandwidth rises. It does not make Alchip the owner of the cited SerDes IP, establish 448G volume production, or award it a new optical component business. September's AI Infra announcement similarly distinguishes 3nm designs in production from 2nm designs in development. [Synopsys224G; AI_Infra]

Alchip emphasizes production management, including work-in-process visibility, cycle-time optimization and yield improvement. Its February release cites experience with packages up to 70x80mm. Those capabilities can reduce a customer's costly waiting time and support better terms for scarce qualified engineering. The model tests that pricing mechanism on engineering work eligible for new terms, rather than applying a premium to every dollar of wafer or memory content. [production_management]

The constraints are real. The annual report identifies TSMC as the principal wafer supplier and says no long-term wafer supply contract is in place. The largest unnamed supplier represented 99.06% of 2025 purchases. TSMC capacity planning visibility and close engineering relationships improve execution, but do not turn nominal industry capacity into Alchip's contracted good-unit output. HBM, substrates, assembly and test must all arrive in the correct sequence. [FY2025_annual, pp.111-112,126]



Original issuer exhibit | Q2 advanced-node mix is not a customer-by-customer shipment or yield disclosure.

The production ramp has moved beyond a promise

Management dates the start of the important North American N3 accelerator shipments to May 2026 and expects the ramp to strengthen through Q3. Official MOPS disclosure provides a tangible check: July sales were NT$7.433 billion and August sales NT$8.776 billion. August rose 18.06% month on month and 273.61% year on year. Those two months alone total NT$16.209 billion, against NT$7.640 billion for the entire second quarter. [Q2_2026_deck, p.9; MOPS_Aug]

The Q3 reference therefore starts with actual July/August sales, then assumes NT$9.1 billion for September. At the model's constant NT$31.85 per USD, that gives approximately USD794.6 million of quarterly sales. A September range of NT$8.0-10.5 billion moves Q3 to roughly USD760-839 million. September and Q3 results were not yet available in the reviewed sources. Q4 revenue of USD1.040 billion is a forward assumption, not a published numerical company forecast.

The fixed exchange rate is a modeling convention anchored to the June balance-sheet conversion. It does not predict the currency. Monthly cumulative figures and financial-statement translations can differ, so the model preserves the disclosed July/August amounts rather than forcing every cumulative series to reconcile mechanically.



Original issuer exhibit | Management dates the North American N3 shipment start to May and identifies growing engineering costs.

Revenue growth and gross profit are different questions

Q2 revenue increased 82.6% sequentially to USD241.728 million. Gross profit rose to USD84.238 million from about USD66.4 million, while gross margin fell from roughly 50.2% to 34.8%. Operating profit increased to USD52.108 million despite operating expenses rising to USD32.130 million. Management attributed that cost increase to headcount, compensation, EDA tools and server usage. This is the expected direction when high-margin design activity gives way to a much larger production mix: percentage margins can fall while absolute profit improves. [Q2_2026_deck, pp.5,9]

The forecast assigns 55% gross margin to the engineering-rich economic bucket, 13% to core North American production and 18% to other production. These are testable assumptions, not disclosed program margins. The 2027 reference contains USD550 million, USD3.2 billion and USD400 million respectively. R&D is USD132 million and selling/administrative expense USD53 million. Costs retain stock compensation, depreciation, software and cloud usage; the model does not add them back to operating profit. Research spending rises with node complexity through 2030 rather than remaining a fixed token percentage of inflated production revenue.

The high-level allocation cannot be uniquely solved from reported statements. Its calibration is to observed consolidated profit and the documented shift toward manufacturing, not a claim that a specific contract earns 55% or 13%. A two-percentage-point movement in a USD4.5 billion production pool changes annual gross profit by USD90 million before offsetting costs. That margin sensitivity can matter more than another large revenue headline.

One historical presentation issue remains explicit: the original Q1 deck showed HPC at 70%, while the Q2 comparative and statutory statements imply about 51% for Q1. The analysis uses the newer statutory classification and does not manufacture a demand change across inconsistent categories. [Q1_2026_deck, p.6; Q2_2026_financials, pp.41-42]

2Q26 Quarterly Income Statement

US$'000	1Q26	2Q26	QoQ (%)	YoY (%)
Revenue	132,366	241,728	82.6	-18.7
COGs	65,948	157,490	138.8	-33.2
Gross Profit	66,417	84,238	26.8	36.7
Op. Exp	23,181	32,130	38.6	33.5
Op. Income	43,236	52,108	20.5	38.7
Non. Op.	13,165	12,288	-6.7	-14.7
Pre-Tax Profit	56,402	64,396	14.2	23.9
Income Tax	-11,284	-12,563	11.3	39.0
Net income	45,118	51,833	14.9	20.7
EPS (NT$)	17.6	20.0		



Original issuer exhibit | Q2 production growth diluted gross margin while raising gross-profit dollars. The deck EPS is basic; the financial statements separately report diluted EPS.

Timing, probability and the next generation

The company's N2 accelerator tape-out expectation is year-end 2026. Tape-out precedes fabrication, testing, package qualification, customer acceptance and sustained volume. The reference includes ongoing engineering work and a gradually larger production business; it does not equate a tape-out with immediate sales. [Q1_2026_deck, p.10]

An October 3 post contains a chart attributed to September Morgan Stanley research: Trainium revenue of USD1.8 billion in 2026, USD2.8 billion in 2027 and USD8 billion in 2028, with the final year representing 82% of company sales. The original bank report and its gross/net assumptions were not obtained. The chart is credible forward evidence to evaluate, rather than an instruction to copy its forecast or a verified identification of the issuer's unnamed customer. [MS_CHART]

The active model assigns 30% probability to a larger 2028 production outcome. The reference North American pool is already USD4.5 billion; the conditional central pool is USD7 billion, bounded at USD5.5-8 billion. Only the USD2.5 billion central residual is incremental. At 10% gross profit less 2% support expense, it produces USD200 million of conditional operating profit and USD60 million after weighting. The corresponding weighted range is USD24-84 million. Design work already present in the reference is not added again.

Pricing is a separate, smaller mechanism. A 35% probability applies to a 4% premium on only 25% of engineering-rich revenue, with half deducted for overlap with ordinary pricing and mix. Seventy percent of that residual reaches gross profit and 5% funds support, leaving 65% operating-profit retention. Existing fixed-price arrangements and manufacturing pass-through are excluded. The premium tapers after 2028.

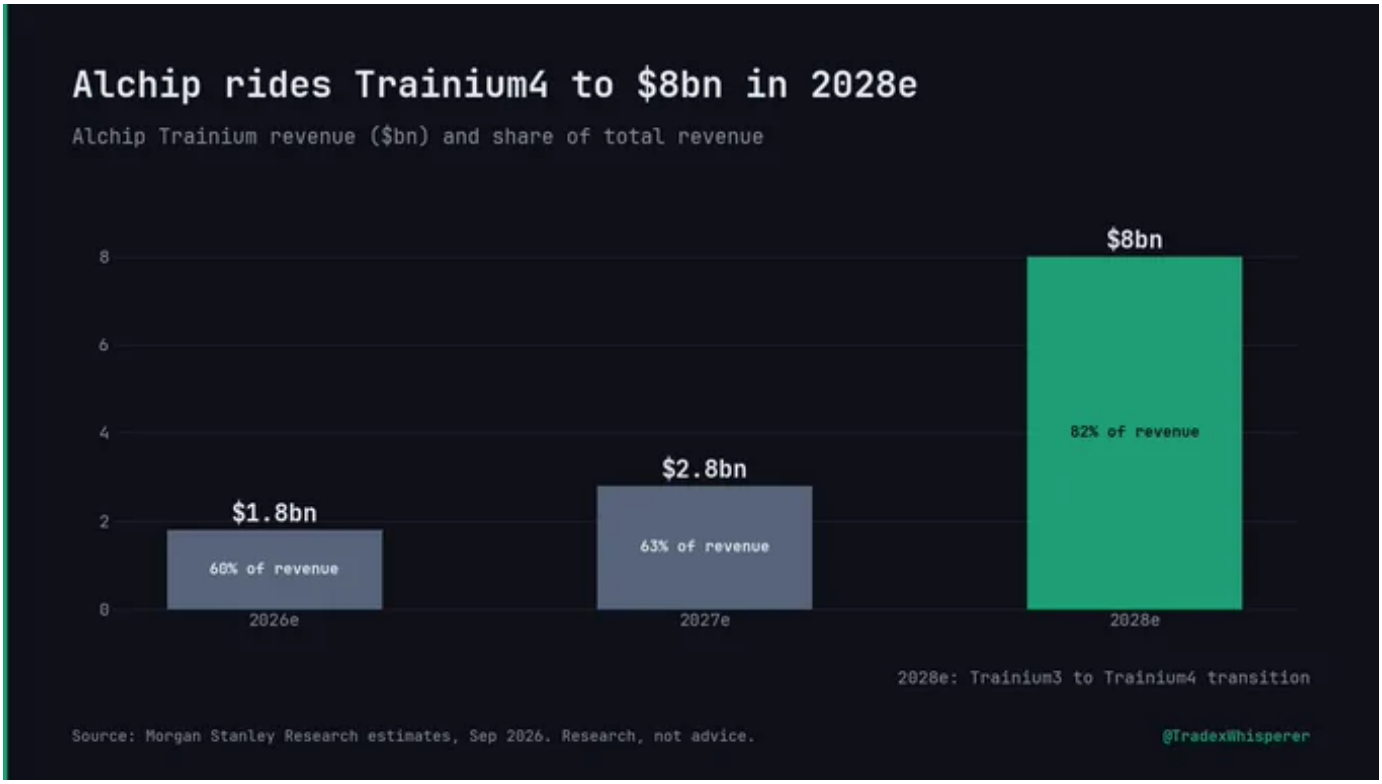
Downside is also explicit. A 25% probability that 20% of the 2027 production pool slips produces a USD160 million weighted revenue reduction and USD17.6 million operating-profit reduction. Half of the delayed amount returns in 2028; the rest is not promised back. An additional 30% probability of USD8 million annual qualification/engineering

cost applies in 2027-28. This recognizes actual extra work without counting wafer, package and customer scheduling constraints as three independent demand shocks.

After those adjustments, modeled revenue is approximately USD3.99 billion in 2027, USD6.47 billion in 2028, USD7.70 billion in 2029 and USD9.01 billion in 2030. Operating profit is approximately USD586 million, USD882 million, USD1.04 billion and USD1.20 billion. These remain scenario-dependent company builds. The issuer's broad USD150 billion ASIC-market aspiration supplies context, not an automatic share-of-TAM revenue formula.



Original issuer exhibit | The issuer expects year-end N2 tape-out; this is not a promise of year-end mass production.



Original issuer exhibit | Attributed September Morgan Stanley estimates, reposted October 3. The USD8B 2028 figure is a conditional scenario source, not Alchip guidance or a copied target.

Working capital is part of the product

June inventory was USD762.403 million, including USD662.147 million of work in progress, versus USD220.982 million of total inventory at year-end. Customer contract liabilities of USD480.275 million and accounts payable of USD244.389 million finance part of the cycle. Even so, H1 operating cash flow was negative USD122.196 million. Accounting profit did not eliminate the need to fund manufacturing. [Q2_2026_financials, pp.15-16,29]

Cash and deposits totaled approximately USD1.675 billion, but USD350.958 million of deposits were pledged. The filing also disclosed USD700 million of raw-material-related guarantee commitments. Pledged money is not a free dividend, and the guarantee face value is not automatically an additional USD700 million of debt. The model instead rolls operating working capital, preserves collateral and reserves USD250 million for ordinary liquidity before adding excess treasury to equity value. Customer advances enter that working-capital bridge once; they are not subtracted again as financial debt. [Q2_2026_financials, pp.24,60]

Future net working capital is modeled at 20% of annual sales, with quarterly transition, and collateral at the greater of the current pledge or 15% of estimated annual production cost. Both percentages need validation against future filings. Cash capital spending, IP investment, dividends and lease claims are retained. The USD8 million annual cloud minimum is inside modeled R&D rather than capitalized as an additional debt claim on top of its operating cost.

（承前頁）

	115年4月1日至6月30日	114年4月1日至6月30日	115年1月1日至6月30日	114年1月1日至6月30日
美　金				
其他員工福利	$ 2,771	$ 3,087	$ 5,375	$ 6,206
股份基礎給付	258	8	414	96
退職後福利	16	7	25	14
	$ 3,045	$ 3,102	$ 5,814	$ 6,316

董事及其他主要管理階層之薪酬係由薪酬委員會依照個人績效並考量市場趨勢決定。

二八、質抵押之資產

截至115年6月30日暨114年12月31日及6月30日止，合併公司分別將定期存款11,178,011仟元（美金350,958仟元）及30,349仟元（美金966仟元）及30,281仟元（美金1,033仟元）作為海關進口原物料及銀行開立擔保信用狀之擔保品。

二九、重大或有負債及未認列之合約承諾

除已於其他附註揭露者外，合併公司於報導期間結束日未認列之合約承諾如下：

(一) 依合併公司與若干公司簽訂矽智財授權合約之規定，合併公司應支付約定金額以取得其於一定期間之授權使用量。其中，與某公司簽定之授權合約，合併公司應依合約協議之付款期間按期支付授權金，截至115年6月30日暨114年12月31日及6月30日止，合併公司尚應分別支付美金8,333仟元、12,550仟元及16,717仟元以獲得相關授權。

(二) 合併公司與甲公司於114年12月31日簽訂雲端運算服務協議，合約期間自115年1月1日至119年12月31日止。合併公司應依實際使用之運算小時數支付費用，並承諾每年度最低使用金額為美金8,000仟元。

(三) 截至115年6月30日止，合併公司因採購原料需求，委請金融機構開立之擔保信用狀相關付款保證金額為美金700,000仟元。

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Original issuer exhibit | Original Chinese note: USD350.958M deposits pledged; USD700M raw-material guarantee commitments; annual cloud minimum USD8M.

Shares, financing and the treasury bridge

June issued shares were approximately 86.205 million, including the new four-million-share GDS offering and a 0.557-million-share private placement. Each GDS represents one ordinary share; it is not an additional layer of common equity beyond the underlying shares. The GDS raised USD510.040 million gross and USD504.131 million net, and the private placement raised USD75 million. Those proceeds and shares are already in the June starting balance. [GDS_offering; Q2_2026_financials, pp.35-39]

The diluted-share footnote needs clarification: Q2 weighted basic shares were 81.892 million and diluted shares 87.512 million, while the separate option note lists only 1.496 million outstanding options. The unexplained 5.586-million potential-option line should not be blindly added to the new issued count. The forecast uses a labeled conservative proxy of 87.766 million: issued shares plus all outstanding options and a 0.065-million employee-compensation reserve, without an assumed option-exercise cash benefit. Future grants add 0.1 million shares annually.

There is also authorization for up to two million new private-placement shares, without disclosed execution terms. A 25% financing state assumes issuance in Q2 2027 at NT$3,000 with 2% fees. It adds 0.5 million weighted

shares and NT$1.470 billion weighted net cash together. The issue-price sensitivity is NT$2,500-4,500. This is an explicit potential funding event, not an announced placement. The valuation uses the average diluted shares across its four operating quarters so it cannot capitalize future profit on a stale pre-raise denominator.

二三、每股盈餘

單位：每股元

	115年4月1日至6月30日	114年4月1日至6月30日	115年1月1日至6月30日	114年1月1日至6月30日
新台幣				
基本每股盈餘	$ 20.01	$ 16.37	$ 37.57	$ 34.49
稀釋每股盈餘	$ 18.72	$ 16.02	$ 35.05	$ 33.71
美金				
基本每股盈餘	$ 0.63	$ 0.53	$ 1.19	$ 1.08
稀釋每股盈餘	$ 0.59	$ 0.52	$ 1.11	$ 1.06

用以計算每股盈餘之資訊如下：

本期淨利

	115年4月1日至6月30日	114年4月1日至6月30日	115年1月1日至6月30日	114年1月1日至6月30日
新台幣				
歸屬於本公司業主之淨利	$ 1,638,567	$ 1,323,522	$ 3,066,460	$ 2,787,040
用以計算基本及稀釋每股盈餘之淨利	$ 1,638,567	$ 1,323,522	$ 3,066,460	$ 2,787,040
美金				
歸屬於本公司業主之淨利	$ 51,853	$ 42,989	$ 96,996	$ 87,480
用以計算基本及稀釋每股盈餘之淨利	$ 51,853	$ 42,989	$ 96,996	$ 87,480

股數

單位：仟股

	115年4月1日至6月30日	114年4月1日至6月30日	115年1月1日至6月30日	114年1月1日至6月30日
用以計算基本每股盈餘之普通股加權平均股數	81,892	80,869	81,617	80,805
具稀釋作用潛在普通股之影響：				
員工認股權	5,586	1,680	5,794	1,792
員工酬勞	34	43	65	73
用以計算稀釋每股盈餘之普通股加權平均股數	87,512	82,592	87,476	82,670

若合併公司得選擇以股票或現金發放員工酬勞，則計算稀釋每股盈餘時，假設員工酬勞將採發放股票方式，並於該潛在普通股具有稀釋作用時計入加權平均流通在外股數，以計算稀釋每股盈餘。於次年度決議員工酬勞發放股數前計算稀釋每股盈餘時，亦繼續考量該等潛在普通股之稀釋作用。

Original issuer exhibit | The reviewed statement reports Q2 basic 81.892M and diluted 87.512M weighted shares. The option line requires reconciliation with the separate option note.

What would change the conclusion

The most valuable next evidence is September/Q3 sales and gross profit, the N2 tape-out and qualification calendar, manufacturing cash conversion, and disclosure clarifying customer principal/agent economics and the diluted-share discrepancy. Strong revenue accompanied by rising pledged deposits and slower inventory turnover is different from clean fee growth. Conversely, stable profit per production dollar with better cash conversion would strengthen both the operating forecast and the treasury contribution.

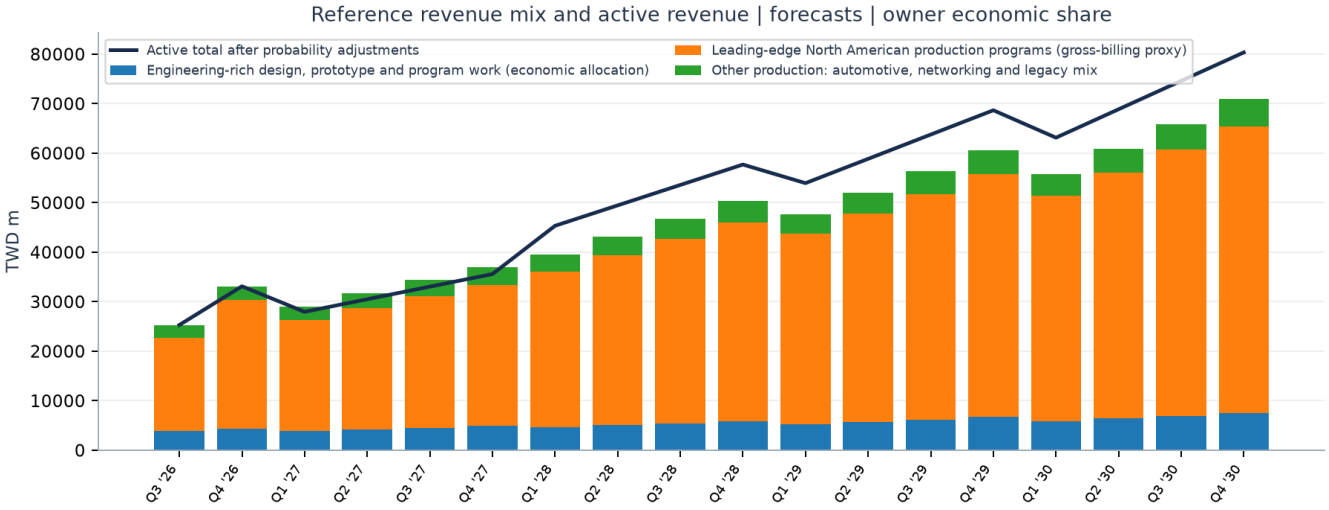
Competition includes customers expanding internal design teams, alternative implementation partners, and suppliers offering more integrated custom silicon. Alchip can preserve value where qualification knowledge and execution shorten time to usable compute; it can lose value if the customer retains more engineering or sources manufacturing directly. In 2025 its two largest customers represented 50.14% and 18.04% of revenue. That concentration, together with the single-supplier purchase exposure, prevents a high growth rate from being

mistaken for diversification. The model's Watch score reflects those economics even though the underlying engineering position is attractive.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	16	See the scoring evidence and distinct risk classes in the analysis.
Cycle	9	See the scoring evidence and distinct risk classes in the analysis.
TAM	9	See the scoring evidence and distinct risk classes in the analysis.
Catalysts	4	See the scoring evidence and distinct risk classes in the analysis.
Macro	7	See the scoring evidence and distinct risk classes in the analysis.
Alpha	3	See the scoring evidence and distinct risk classes in the analysis.
Risk classes	-4	See the scoring evidence and distinct risk classes in the analysis.
Total	44/60	Rounded Base 73. Both artifact and new workbook row use this score; this is not a manual override.

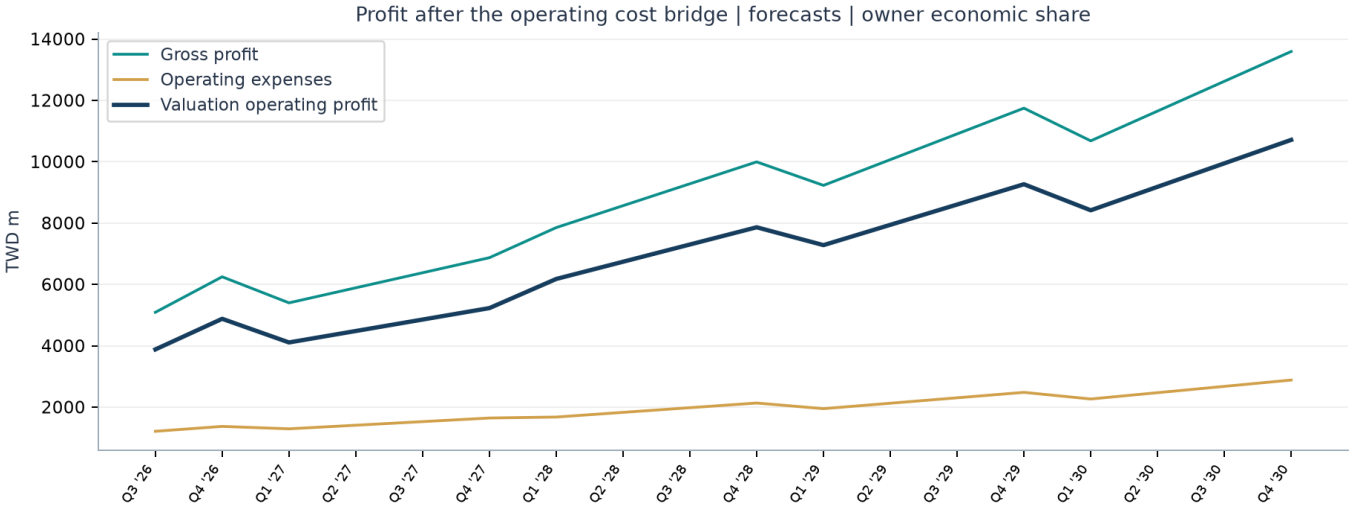
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	25,309.4	0.0	25,309.4
Q4 2026	33,124.0	0.0	33,124.0
Q1 2027	29,079.1	-1,114.4	27,964.7
Q2 2027	31,722.6	-1,215.7	30,506.9
Q3 2027	34,366.2	-1,317.0	33,049.2
Q4 2027	37,009.7	-1,418.3	35,591.4
Q1 2028	39,519.5	5,823.9	45,343.4
Q2 2028	43,112.2	6,353.3	49,465.5
Q3 2028	46,704.8	6,882.8	53,587.6
Q4 2028	50,297.5	7,412.2	57,709.8
Q1 2029	47,647.6	6,310.9	53,958.5
Q2 2029	51,979.2	6,884.6	58,863.8
Q3 2029	56,310.8	7,458.3	63,769.1
Q4 2029	60,642.4	8,032.1	68,674.5
Q1 2030	55,775.7	7,362.5	63,138.2
Q2 2030	60,846.2	8,031.8	68,878.1
Q3 2030	65,916.8	8,701.1	74,617.9
Q4 2030	70,987.3	9,370.5	80,357.7

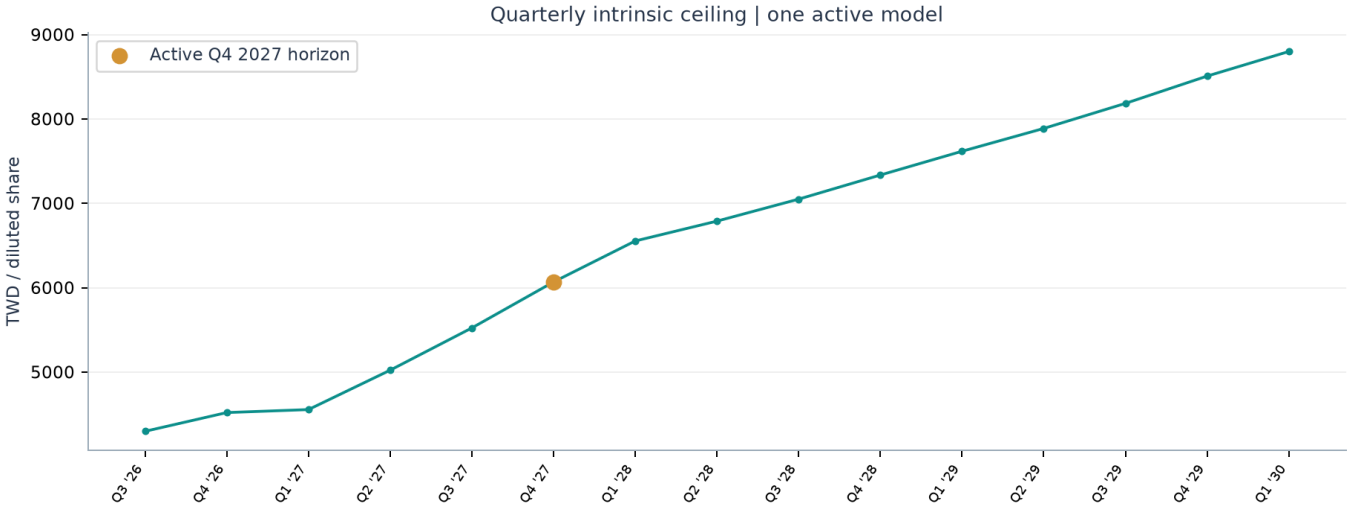
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	25,309.4	5,089.8	1,210.3	3,879.5	15.3%
Q4 2026	33,124.0	6,247.4	1,369.6	4,877.8	14.7%
Q1 2027	27,964.7	5,398.0	1,291.0	4,107.0	14.7%
Q2 2027	30,506.9	5,888.7	1,408.4	4,480.3	14.7%
Q3 2027	33,049.2	6,379.5	1,525.8	4,853.7	14.7%
Q4 2027	35,591.4	6,870.2	1,643.1	5,227.1	14.7%
Q1 2028	45,343.4	7,852.1	1,675.1	6,177.0	13.6%
Q2 2028	49,465.5	8,565.9	1,827.4	6,738.6	13.6%
Q3 2028	53,587.6	9,279.8	1,979.6	7,300.1	13.6%
Q4 2028	57,709.8	9,993.6	2,131.9	7,861.7	13.6%
Q1 2029	53,958.5	9,227.9	1,948.2	7,279.8	13.5%
Q2 2029	58,863.8	10,066.8	2,125.3	7,941.6	13.5%
Q3 2029	63,769.1	10,905.7	2,302.4	8,603.4	13.5%
Q4 2029	68,674.5	11,744.6	2,479.5	9,265.1	13.5%
Q1 2030	63,138.2	10,679.5	2,263.5	8,416.0	13.3%
Q2 2030	68,878.1	11,650.3	2,469.3	9,181.0	13.3%
Q3 2030	74,617.9	12,621.2	2,675.1	9,946.1	13.3%
Q4 2030	80,357.7	13,592.1	2,880.8	10,711.2	13.3%

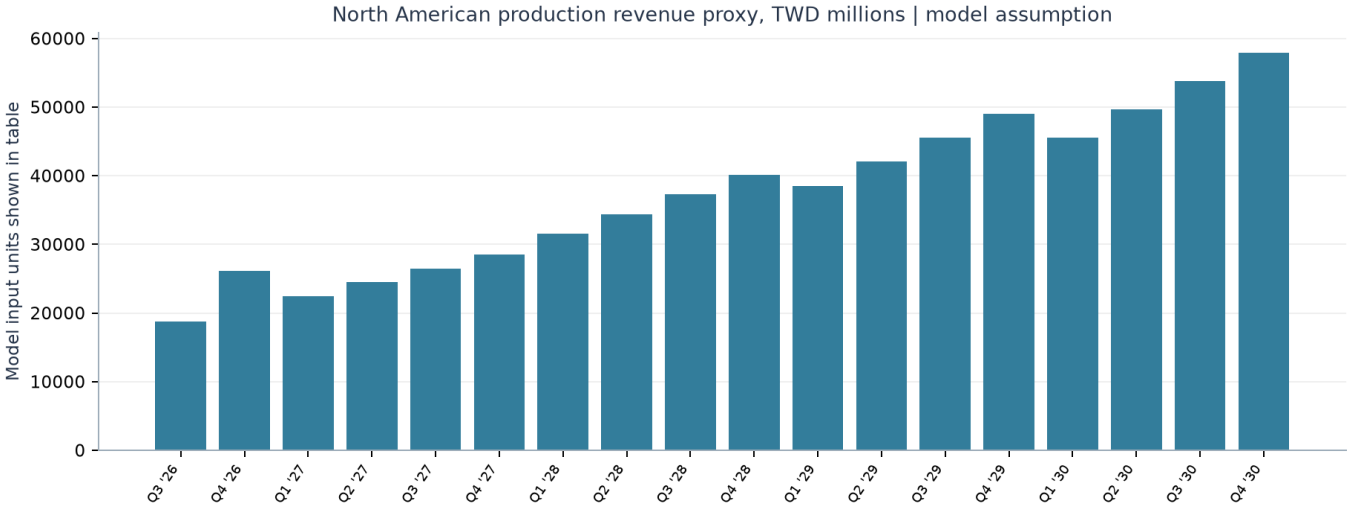
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Exact earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 + Q4 2026 + Q1 2027 + Q2 2027	17,344.6	20x	87.929	4,297.67
Q4 2026	Q4 2026 + Q1 2027 + Q2 2027 + Q3 2027	18,318.9	20x	88.079	4,519.63
Q1 2027	Q1 2027 + Q2 2027 + Q3 2027 + Q4 2027	18,668.1	20x	88.228	4,555.08
Q2 2027	Q2 2027 + Q3 2027 + Q4 2027 + Q1 2028	20,738.1	20x	88.379	5,021.99
Q3 2027	Q3 2027 + Q4 2027 + Q1 2028 + Q2 2028	22,996.4	20x	88.404	5,522.89
Q4 2027	Q4 2027 + Q1 2028 + Q2 2028 + Q3 2028	25,442.8	20x	88.429	6,067.92
Q1 2028	Q1 2028 + Q2 2028 + Q3 2028 + Q4 2028	28,077.4	20x	88.454	6,553.71
Q2 2028	Q2 2028 + Q3 2028 + Q4 2028 + Q1 2029	29,180.1	20x	88.478	6,788.96
Q3 2028	Q3 2028 + Q4 2028 + Q1 2029 + Q2 2029	30,383.1	20x	88.504	7,049.52
Q4 2028	Q4 2028 + Q1 2029 + Q2 2029 + Q3 2029	31,686.3	20x	88.529	7,335.37
Q1 2029	Q1 2029 + Q2 2029 + Q3 2029 + Q4 2029	33,089.8	20x	88.554	7,617.00
Q2 2029	Q2 2029 + Q3 2029 + Q4 2029 + Q1 2030	34,226.0	20x	88.579	7,887.57
Q3 2029	Q3 2029 + Q4 2029 + Q1 2030 + Q2 2030	35,465.5	20x	88.603	8,184.58
Q4 2029	Q4 2029 + Q1 2030 + Q2 2030 + Q3 2030	36,808.3	20x	88.629	8,508.02
Q1 2030	Q1 2030 + Q2 2030 + Q3 2030 + Q4 2030	38,254.3	20x	88.654	8,799.56

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	North American production revenue proxy, TWD millions	Status
Q3 2026	18,780.188	Observed July/August plus September assumption
Q4 2026	26,117.000	Labeled operating assumptions
Q1 2027	22,422.400	Labeled operating assumptions
Q2 2027	24,460.800	Labeled operating assumptions
Q3 2027	26,499.200	Labeled operating assumptions
Q4 2027	28,537.600	Labeled operating assumptions
Q1 2028	31,531.500	Labeled operating assumptions
Q2 2028	34,398.000	Labeled operating assumptions
Q3 2028	37,264.500	Labeled operating assumptions
Q4 2028	40,131.000	Labeled operating assumptions
Q1 2029	38,538.500	Labeled operating assumptions
Q2 2029	42,042.000	Labeled operating assumptions
Q3 2029	45,545.500	Labeled operating assumptions
Q4 2029	49,049.000	Labeled operating assumptions
Q1 2030	45,545.500	Labeled operating assumptions
Q2 2030	49,686.000	Labeled operating assumptions
Q3 2030	53,826.500	Labeled operating assumptions
Q4 2030	57,967.000	Labeled operating assumptions

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Engineering-rich design, prototype and program work (economic allocation)	3,981.2	55.0%	2,189.7
Q3 2026	Leading-edge North American production programs (gross-billing proxy)	18,780.2	13.0%	2,441.4
Q3 2026	Other production: automotive, networking and legacy mix	2,548.0	18.0%	458.6
Q4 2026	Engineering-rich design, prototype and program work (economic allocation)	4,299.8	55.0%	2,364.9
Q4 2026	Leading-edge North American production programs (gross-billing proxy)	26,117.0	13.0%	3,395.2
Q4 2026	Other production: automotive, networking and legacy mix	2,707.2	18.0%	487.3
Q1 2027	Engineering-rich design, prototype and program work (economic allocation)	3,853.9	55.0%	2,119.6
Q1 2027	Leading-edge North American production programs (gross-billing proxy)	22,422.4	13.0%	2,914.9
Q1 2027	Other production: automotive, networking and legacy mix	2,802.8	18.0%	504.5
Q2 2027	Engineering-rich design, prototype and program work (economic allocation)	4,204.2	55.0%	2,312.3
Q2 2027	Leading-edge North American production programs (gross-billing proxy)	24,460.8	13.0%	3,179.9
Q2 2027	Other production: automotive, networking and legacy mix	3,057.6	18.0%	550.4
Q3 2027	Engineering-rich design, prototype and program work (economic allocation)	4,554.6	55.0%	2,505.0
Q3 2027	Leading-edge North American production programs (gross-billing proxy)	26,499.2	13.0%	3,444.9
Q3 2027	Other production: automotive, networking and legacy mix	3,312.4	18.0%	596.2
Q4 2027	Engineering-rich design, prototype and program work (economic allocation)	4,904.9	55.0%	2,697.7
Q4 2027	Leading-edge North American production programs (gross-billing proxy)	28,537.6	13.0%	3,709.9
Q4 2027	Other production: automotive, networking and legacy mix	3,567.2	18.0%	642.1
Q1 2028	Engineering-rich design, prototype and program work (economic allocation)	4,624.6	55.0%	2,543.5
Q1 2028	Leading-edge North American production programs (gross-billing proxy)	31,531.5	13.0%	4,099.1

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2028	Other production: automotive, networking and legacy mix	3,363.4	18.0%	605.4
Q2 2028	Engineering-rich design, prototype and program work (economic allocation)	5,045.0	55.0%	2,774.8
Q2 2028	Leading-edge North American production programs (gross-billing proxy)	34,398.0	13.0%	4,471.7
Q2 2028	Other production: automotive, networking and legacy mix	3,669.1	18.0%	660.4
Q3 2028	Engineering-rich design, prototype and program work (economic allocation)	5,465.5	55.0%	3,006.0
Q3 2028	Leading-edge North American production programs (gross-billing proxy)	37,264.5	13.0%	4,844.4
Q3 2028	Other production: automotive, networking and legacy mix	3,974.9	18.0%	715.5
Q4 2028	Engineering-rich design, prototype and program work (economic allocation)	5,885.9	55.0%	3,237.2
Q4 2028	Leading-edge North American production programs (gross-billing proxy)	40,131.0	13.0%	5,217.0
Q4 2028	Other production: automotive, networking and legacy mix	4,280.6	18.0%	770.5
Q1 2029	Engineering-rich design, prototype and program work (economic allocation)	5,255.2	55.0%	2,890.4
Q1 2029	Leading-edge North American production programs (gross-billing proxy)	38,538.5	13.0%	5,010.0
Q1 2029	Other production: automotive, networking and legacy mix	3,853.9	18.0%	693.7
Q2 2029	Engineering-rich design, prototype and program work (economic allocation)	5,733.0	55.0%	3,153.2
Q2 2029	Leading-edge North American production programs (gross-billing proxy)	42,042.0	13.0%	5,465.5
Q2 2029	Other production: automotive, networking and legacy mix	4,204.2	18.0%	756.8
Q3 2029	Engineering-rich design, prototype and program work (economic allocation)	6,210.8	55.0%	3,415.9
Q3 2029	Leading-edge North American production programs (gross-billing proxy)	45,545.5	13.0%	5,920.9
Q3 2029	Other production: automotive, networking and legacy mix	4,554.6	18.0%	819.8
Q4 2029	Engineering-rich design, prototype and program work (economic allocation)	6,688.5	55.0%	3,678.7
Q4 2029	Leading-edge North American production programs (gross-billing proxy)	49,049.0	13.0%	6,376.4
Q4 2029	Other production: automotive, networking and legacy mix	4,904.9	18.0%	882.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2030	Engineering-rich design, prototype and program work (economic allocation)	5,885.9	55.0%	3,237.2
Q1 2030	Leading-edge North American production programs (gross-billing proxy)	45,545.5	13.0%	5,920.9
Q1 2030	Other production: automotive, networking and legacy mix	4,344.3	18.0%	782.0
Q2 2030	Engineering-rich design, prototype and program work (economic allocation)	6,421.0	55.0%	3,531.5
Q2 2030	Leading-edge North American production programs (gross-billing proxy)	49,686.0	13.0%	6,459.2
Q2 2030	Other production: automotive, networking and legacy mix	4,739.3	18.0%	853.1
Q3 2030	Engineering-rich design, prototype and program work (economic allocation)	6,956.0	55.0%	3,825.8
Q3 2030	Leading-edge North American production programs (gross-billing proxy)	53,826.5	13.0%	6,997.4
Q3 2030	Other production: automotive, networking and legacy mix	5,134.2	18.0%	924.2
Q4 2030	Engineering-rich design, prototype and program work (economic allocation)	7,491.1	55.0%	4,120.1
Q4 2030	Leading-edge North American production programs (gross-billing proxy)	57,967.0	13.0%	7,535.7
Q4 2030	Other production: automotive, networking and legacy mix	5,529.2	18.0%	995.2

Operating-cost derivation

Quarter	R&D including EDA, servers, employee costs and SBC	Selling and administrative expenses including SBC
Q3 2026	860.0	350.4
Q4 2026	987.4	382.2
Q1 2027	924.9	371.4
Q2 2027	1,009.0	405.1
Q3 2027	1,093.1	438.9
Q4 2027	1,177.2	472.7
Q1 2028	1,121.1	420.4
Q2 2028	1,223.0	458.6
Q3 2028	1,325.0	496.9
Q4 2028	1,426.9	535.1
Q1 2029	1,331.3	490.5
Q2 2029	1,452.4	535.1
Q3 2029	1,573.4	579.7
Q4 2029	1,694.4	624.3
Q1 2030	1,555.6	560.6
Q2 2030	1,697.0	611.5
Q3 2030	1,838.4	662.5
Q4 2030	1,979.8	713.4

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q3 2026	ALCHIP-REPRICE	35%	0.0	0.0	0.0	0.0
Q3 2026	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q3 2026	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q3 2026	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q4 2026	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q4 2026	ALCHIP-REPRICE	35%	0.0	0.0	0.0	0.0
Q4 2026	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q4 2026	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q4 2026	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q1 2027	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q1 2027	ALCHIP-REPRICE	35%	19.3	13.5	1.0	4.4
Q1 2027	ALCHIP-DELAY	25%	-4,484.5	-583.0	-89.7	-123.3
Q1 2027	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	56.1	-16.8
Q1 2027	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q2 2027	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q2 2027	ALCHIP-REPRICE	35%	21.0	14.7	1.1	4.8
Q2 2027	ALCHIP-DELAY	25%	-4,892.2	-636.0	-97.8	-134.5
Q2 2027	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	61.2	-18.3
Q2 2027	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q3 2027	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q3 2027	ALCHIP-REPRICE	35%	22.8	15.9	1.1	5.2
Q3 2027	ALCHIP-DELAY	25%	-5,299.8	-689.0	-106.0	-145.7
Q3 2027	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	66.2	-19.9
Q3 2027	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q4 2027	ALCHIP-NEXT-GENERATION	30%	0.0	0.0	0.0	0.0
Q4 2027	ALCHIP-REPRICE	35%	24.5	17.2	1.2	5.6
Q4 2027	ALCHIP-DELAY	25%	-5,707.5	-742.0	-114.2	-157.0
Q4 2027	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	71.3	-21.4
Q4 2027	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q1 2028	ALCHIP-NEXT-GENERATION	30%	17,517.5	1,751.8	350.4	420.4
Q1 2028	ALCHIP-REPRICE	35%	23.1	16.2	1.2	5.3
Q1 2028	ALCHIP-DELAY	25%	2,242.2	291.5	44.8	61.7
Q1 2028	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	56.1	-16.8
Q1 2028	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q2 2028	ALCHIP-NEXT-GENERATION	30%	19,110.0	1,911.0	382.2	458.6
Q2 2028	ALCHIP-REPRICE	35%	25.2	17.7	1.3	5.7
Q2 2028	ALCHIP-DELAY	25%	2,446.1	318.0	48.9	67.3
Q2 2028	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	61.2	-18.3
Q2 2028	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q3 2028	ALCHIP-NEXT-GENERATION	30%	20,702.5	2,070.2	414.1	496.9
Q3 2028	ALCHIP-REPRICE	35%	27.3	19.1	1.4	6.2
Q3 2028	ALCHIP-DELAY	25%	2,649.9	344.5	53.0	72.9
Q3 2028	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	66.2	-19.9
Q3 2028	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q4 2028	ALCHIP-NEXT-GENERATION	30%	22,295.0	2,229.5	445.9	535.1
Q4 2028	ALCHIP-REPRICE	35%	29.4	20.6	1.5	6.7
Q4 2028	ALCHIP-DELAY	25%	2,853.8	371.0	57.1	78.5
Q4 2028	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	71.3	-21.4
Q4 2028	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q1 2029	ALCHIP-NEXT-GENERATION	30%	21,021.0	2,102.1	420.4	504.5
Q1 2029	ALCHIP-REPRICE	35%	13.1	9.2	0.7	3.0

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q1 2029	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q1 2029	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q1 2029	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q2 2029	ALCHIP-NEXT-GENERATION	30%	22,932.0	2,293.2	458.6	550.4
Q2 2029	ALCHIP-REPRICE	35%	14.3	10.0	0.7	3.3
Q2 2029	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q2 2029	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q2 2029	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q3 2029	ALCHIP-NEXT-GENERATION	30%	24,843.0	2,484.3	496.9	596.2
Q3 2029	ALCHIP-REPRICE	35%	15.5	10.9	0.8	3.5
Q3 2029	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q3 2029	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q3 2029	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q4 2029	ALCHIP-NEXT-GENERATION	30%	26,754.0	2,675.4	535.1	642.1
Q4 2029	ALCHIP-REPRICE	35%	16.7	11.7	0.8	3.8
Q4 2029	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q4 2029	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q4 2029	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q1 2030	ALCHIP-NEXT-GENERATION	30%	24,524.5	2,452.5	490.5	588.6
Q1 2030	ALCHIP-REPRICE	35%	14.7	10.3	0.7	3.3
Q1 2030	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q1 2030	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q1 2030	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q2 2030	ALCHIP-NEXT-GENERATION	30%	26,754.0	2,675.4	535.1	642.1
Q2 2030	ALCHIP-REPRICE	35%	16.1	11.2	0.8	3.7
Q2 2030	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q2 2030	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q2 2030	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q3 2030	ALCHIP-NEXT-GENERATION	30%	28,983.5	2,898.3	579.7	695.6
Q3 2030	ALCHIP-REPRICE	35%	17.4	12.2	0.9	4.0
Q3 2030	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q3 2030	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q3 2030	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0
Q4 2030	ALCHIP-NEXT-GENERATION	30%	31,213.0	3,121.3	624.3	749.1
Q4 2030	ALCHIP-REPRICE	35%	18.7	13.1	0.9	4.3
Q4 2030	ALCHIP-DELAY	25%	0.0	0.0	0.0	0.0
Q4 2030	ALCHIP-QUALIFICATION-COST	30%	0.0	0.0	0.0	0.0
Q4 2030	ALCHIP-FINANCING	25%	0.0	0.0	0.0	0.0

Enterprise-to-equity reconciliation

20x forward-four-quarter modeled IFRS operating profit, plus only excess treasury after collateral/operating reserves, less lease and minority claims, divided by matched forward diluted shares

Quarter	Reference claims	Weighted change	Total claims	Ceiling / share
Q3 2026	-30,995.2	0.0	-30,995.2	4,297.67
Q4 2026	-31,705.1	0.0	-31,705.1	4,519.63
Q1 2027	-27,803.4	-722.4	-28,525.8	4,555.08
Q2 2027	-26,692.7	-2,380.5	-29,073.2	5,021.99
Q3 2027	-25,751.8	-2,563.1	-28,314.9	5,522.89
Q4 2027	-24,980.5	-2,740.4	-27,720.9	6,067.92
Q1 2028	-20,072.5	1,922.0	-18,150.5	6,553.71
Q2 2028	-20,347.2	3,273.0	-17,074.2	6,788.96
Q3 2028	-20,850.2	4,605.2	-16,245.0	7,049.52
Q4 2028	-21,581.5	5,918.6	-15,662.8	7,335.37
Q1 2029	-18,753.3	6,037.1	-12,716.2	7,617.00
Q2 2029	-20,053.2	5,904.2	-14,149.0	7,887.57
Q3 2029	-21,623.9	5,751.0	-15,872.9	8,184.58
Q4 2029	-23,465.6	5,577.5	-17,888.1	8,508.02
Q1 2030	-21,080.3	6,055.0	-15,025.2	8,799.56
Q2 2030	-22,865.1	6,009.9	-16,855.2	-
Q3 2030	-24,962.8	5,941.1	-19,021.7	-
Q4 2030	-27,373.6	5,848.6	-21,525.0	-

Claim decisions, overlap and triggers

ALCHIP-N3-RAMP

probability: 0.8

hypothesis: The observed 3nm ramp converts to sustained customer-accepted production.

Conditional: Q3 reference USD794.65M approximately, from actual July/August NT$16.209438B plus September NT$9.1B assumption. September bound NT$8.0-10.5B implies Q3 USD760-839M at fixed FX31.85. Q4 reference USD1.040B.

overlap: 100% of May start, July/August sales and current volume ramp is in reference. The 80% assessment is execution confidence, not multiplied again into observed sales; the specific 2027 25% delay branch captures prospective downside.

costs: Production GP13% and engineering-rich program GP55% are allocation assumptions; manufacturing pass-through is never valued at the engineering margin.

source ids: Q2_2026_deck MOPS_Aug

ALCHIP-NEXT-GENERATION

probability: 0.3

hypothesis: N2 and next-generation North American accelerator programs expand beyond the reference ramp.

Conditional: 2028 reference North American production USD4.5B; conditional USD7B with a bounded USD5.5-8B outcome. Central residual USD2.5B at 10% GP less 2% support costs becomes USD60M probability-weighted OP. 2029/2030 conditional gross-billing pools USD8.5B/USD10B versus reference USD5.5B/USD6.5B.

overlap: Subtract the full reference production pool first. N2 design fees and ordinary engineering growth are already in the engineering vector. The MS chart’s USD8B is an attributed revenue scenario, not guidance or proof of a named award; its 2026/27 amounts create no duplicate overlay.

bounds: 2028 conditional incremental OP USD80-280M for the USD1-3.5B residual at 8% OP; weighted USD24-84M. Timing can move beyond 2028.

costs: 90% of added gross billings consumed by manufacturing/packaging and 2% by engineering/support; funding and collateral grow with expected sales.

source ids: MS_CHART 2nm Q1_2026_deck AI_Infra

ALCHIP-REPRICE

probability: 0.35

hypothesis: Scarce qualified design and production engineering secures better new-contract economics.

Conditional: Only 25% of engineering-rich program revenue is repriceable. A 2-6% conditional premium, central 4% through 2028 then 2%, less 50% overlap, leaves an incremental 65% OP contribution after labor/EDA/support costs.

overlap: 50% ordinary price/mix overlap. Wafer, HBM and packaging billings are excluded from the eligible engineering pool. No premium assigned to locked contracts.

costs: 70% incremental GP, 5% support opex; collection only with delivered milestones.

source ids: production_management business_models FY2025_annual

ALCHIP-DELAY

probability: 0.25

hypothesis: Advanced-node qualification, packaging availability or customer scheduling delays 2027 deliveries.

Conditional: 20% of the reference USD3.2B production pool delayed in the conditional case, with 10-30% bound; half of delayed sales recover in2028. Central weighted 2027 revenue-USD160M and OP-USD17.6M.

overlap: One correlated delivery event; wafer, substrate and acceptance constraints are not stacked as separate revenue shocks. The recovery is timing, not a new design win.

costs: 13% lost GP less 2% avoided variable support. Separate 30% x USD8M annual qualification-cost branch for 2027-28 recognizes a distinct engineering burden.

source ids: FY2025_annual Q2_2026_deck

ALCHIP-FINANCING

probability: 0.25

hypothesis: Additional strategic capital is raised under the authorized unissued placement capacity.

Conditional: 2M shares issued in Q2 2027 at assumed NT$3,000 with 2% fees: conditional NT$5,880M net proceeds; weighted 0.5M new shares and NT$1,470M cash. Price bound NT$2,500-4,500; no announced deal or committed terms.

overlap: June 4M GDS and 0.557M private placement are already in 86.205M issued shares andJune cash. They are not added again. Full option dilution uses 87.766M starting proxy, plus 0.1M/year future grants.

costs: Funds and dilution enter in the same quarter. Only unencumbered cash beyond pledged collateral and USD250M operating reserve contributes to equity value.

source ids: Q2_2026_financials GDS_offering

ALCHIP-OPTICS-224G

probability: 0.55

hypothesis: Advanced electrical/optical packaging capability improves competitive qualification.

Conditional: Readiness supports the existing engineering and next-generation production paths. No extra optical-source or SerDes revenue is credited to Alchip.

overlap: 100% of the general capability mechanism is embedded in reference design work and the next-generation branch. Synopsys owns the cited PHY; demonstration does not establish commercial volume.

costs: Ongoing engineering, EDA and cloud usage retained. USD8M annual minimum cloud commitment sits within R&D, not an extra debt deduction.

source ids: Synopsys224G AI_Infra Q2_2026_financials

Remaining bounded uncertainties

- The economic engineering/production split and corresponding gross margins are modeled allocations. Statutory ASIC/wafer sales include activity that management calls NRE; do not equate the USD7.498M H1 standalone design-service line with all engineering economics.

- No current program-level unit volume, chip ASP, wafer allocation, packaging yield, customer acceptance clause or gross/net principal assessment is publicly quantified. The North American customer is not named in the reviewed Q2 issuer guidance.

- Q1 application mix differs between its original deck (HPC 70%) and the Q2 comparative/financial statements (approximately 51%). Use the newer statutory presentation for history; do not manufacture a growth bridge across inconsistent classifications.

- The H1 EPS note lists 5.586M Q2 potential option shares, above 1.496M outstanding options in the option note. The filing does not explain the difference; the 87.766M forward full-dilution proxy is 86.205M issued plus 1.496M options plus 0.065M employee-pay reserve, without treasury-method cash credit.

- The September-attributed MS chart is preserved, but the original research report, program assumptions and gross/net revenue definition remain unavailable. Its 2028 forecast is not a committed customer order.

- September monthly sales and Q3 financial results were not published in the sources reviewed as of October 7. The September NT$9.1B assumption and fixed 31.85 USD/TWD conversion are explicit forecast choices.

- Gross cash is not excess cash. Inventory, client advances, pledged deposits and guarantees require a funding bridge. Future collateral and net working capital percentages are bounded proxies, not disclosed contractual terms.

- The authorized 2M-share placement has no disclosed execution terms. The model includes a 25% financing state with matched cash and shares rather than assuming all authority is issued or wholly ignoring it.

- Company conference-call Q&A recordings were not obtained; full issuer Q1/Q2 presentations, current quarterly statements, annual business/risks and current technical releases were reviewed.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

[1] Alchip Q2 2026 earnings presentation | 2026-08-14

https://www.alchip.com/upload/2026_08_14/3_20260814141602cp238o1pY1.pdf

Archived source SHA-256: 1025e1c67ef48a3aeaedb1226fd8428b7fe87b9bce9a563aaefbd5d116f43e86

[2] Alchip Q1 2026 earnings presentation | 2026-05-08

https://www.alchip.com/upload/2026_05_08/6_202605081411542ml4h2pjq0.pdf

Archived source SHA-256: 38f0c9a23d1f946c605a3c3bc9963401635c14654e41bcba1787858fe280be9f

[3] Alchip Reviewed consolidated H1 2026 financial statements, dual USD/TWD | 2026-08-14

https://www.alchip.com/upload/2026_08_20/3_20260820142747byzwj5qm60.pdf

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[4] Alchip 2025 annual report | 2026-08-18 upload; report covers FY2025

https://www.alchip.com/upload/2026_08_18/3_2026081817300451l64sc83t0.pdf

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[5] Alchip Flexible engagement and hand-off models | Undated, retrieved 2026-10-07

https://www.alchip.com/en/Business_Models/model

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[6] Alchip Production management and advanced packaging capabilities | 2026-02-10

https://www.alchip.com/en/Newsroom/AI_Elevates_Production_Managements_Importance

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[7] Alchip 2nm design-platform development | 2026-03-12

https://www.alchip.com/en/Newsroom/Alchip_Reports_ASIC-Leading_2nm_Developments

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[8] Alchip USD510 million GDS offering | 2026-06-26

https://www.alchip.com/en/Newsroom/Alchip_Successfully_Completes_US510MM_GDS_Offering

Archived source SHA-256: 9eb6f4f3fa59577912d021d85f237e7f169e8028497d736e2cf6a097b3f40f81

[9] Alchip Synopsys 224G PHY and package co-design | 2026-09-23

https://www.alchip.com/en/Newsroom/Alchip_Synopsys_TSMC_OIP

Archived source SHA-256: 7127a59d128b23b6063540720b8930e5e7bc4a8efc64ff2b0b6ddb45632ba363

[10] Alchip 3nm production, 2nm development and optical interconnect capabilities | 2026-09-14

https://www.alchip.com/en/Newsroom/Alchip_AI_Infra_Summit_2026

Archived source SHA-256: 0037e769011fbf2977ca06fc861af438cf1e75c01cebecb581fa0a690e880b46

[11] Alchip Q2 financial results press release | 2026-08-26

https://www.alchip.com/en/Newsroom/Alchip_2026_Q2_financial_results

Archived source SHA-256: bbe53eff799d0a3030329576212acbd7c834dadad9807da21f7b070534f5f43f

[12] Official MOPS August 2026 monthly revenue, targeted Alchip row | 2026-09-10 disclosure; August period

https://mopsov.twse.com.tw/nas/t21/sii/t21sc03_115_8.html

Archived source SHA-256: c5a98d0e8d31c6ad6c1c660798a662f268a000b98c2ba573d9d0c50295201f7e

[13] Trainium3/4 revenue chart attributed to Morgan Stanley September research | 2026-10-03 post; chart attributes September 2026 estimates

https://x.com/TradexWhisperer/status/2106487814913483056

Archived source SHA-256: 31db8aeecce839484ee7fb9e526427f6e7f0753a61eaa12da760303b55f92107

Model formulas and equity bridge: 20x forward-four-quarter modeled IFRS operating profit, plus only excess treasury after collateral/operating reserves, less lease and minority claims, divided by matched forward diluted shares Revenue, gross margins, costs, conditional deltas, financing and share inputs are shown above. The workbook links the active ceiling to the same saved model.